

INTERSECTION NUMBERS AND SPLIT OF MINOR ROOTS

YANSONG XU

ABSTRACT. There are two main parts in this manuscript. First, for a Jacobian pair (f, g) , with the concept of final major roots and final minor roots, we obtain equations and inequalities for the intersection numbers $I(f_\xi, g)$ and $I(f_\xi, f_y)$ respectively, here ξ is a generic element of the base field and $f_\xi = f - \xi$ and discuss usage of them in some special cases. Second, we discuss all possibilities of the splits of principal minor roots for the case of degree (99, 66) with help of Abhyankar-Moh planar semigroup, find an unknown possible split and suggest case (99, 66) is open.

1. INTRODUCTION

Through out this paper, $\mathbb{K}$ denotes an algebraically closed field of characteristic 0.

Let $(f(x, y), g(x, y))$ be a Jacobian pair. The Jacobian condition $f_x g_y - f_y g_x = J$ is an expression of two terms. We take a root of $f(y)f_y(y)$ in the Puiseux field $\mathbb{K} \ll x^{-1} \gg$ and plug in the expression. The expression becomes one term, therefore the orders of the factors of the expression are calculable in conditions.

Moh in [Moh83] first introduces the concepts of π -root, major root and minor root. He gets plentiful results for major roots and a few for minor roots. For minor roots, Moh proves that any minor roots are not split before order 1. We introduce the concept of final roots. We find some relationship between intersection numbers and the split of final minor roots. On the other hand, $I(f_\xi, g)$ can be expressed by only final major roots while $I(f_\xi, f_y)$ can also be expressed by total split π -roots. These formulas and inequalities of final major roots and final minor roots provide some constrains for possible counter examples for Jacobian Conjecture.

With the help of quasi-approximate roots and Abhyankar-Moh planar semi-group theory, we show that for any case, the principal minor roots are not splitting at order 1 for all effective quasi-approximate roots. This provide us a tool to further study the split possibilities for the principal minor roots.

In the last section, we discuss the split possibilities for case (99, 66) that Moh does not discuss in details and find an unknown possible split and suggest case (99, 66) is open.

2. PRELIMINARIES

For polynomials $f(x, y), g(x, y) \in \mathbb{K}[x, y]$, monic in y , we define intersection number of f with g as $I(f, g) = \deg_x \text{Res}_y(f, g)$. Let $x = t^{-1}$. We simply write $f(t^{-1}, y)$ as $f(y)$. $f(y)$ can be factored out completely over Puiseux field $\mathbb{K} \ll t \gg$.

Let π be a symbol. Following [Moh83], we define $\sigma = \sum_{j < \delta_\sigma} a_j t^j + \pi t^{\delta_\sigma}$ as a π -root of f if $f(\sigma) = f_\sigma(\pi) t^{\lambda_\sigma} + \dots$ and $\deg f_\sigma(\pi) > 0$. The multiplicity of σ is defined as $\deg_\pi f_\sigma(\pi)$ [Moh83]. We call σ is split if $f_\sigma(\pi)$ has more than one roots. We call δ_σ the split order of σ , or we say f splits at order δ_σ . We call σ is final if $f_\sigma(\pi)$ has no multiple roots and $\deg_\pi f_\sigma(\pi) > 1$. We also call a final π -root as a final root for simplicity.

2010 *Mathematics Subject Classification.* Primary 14R15.

Key words and phrases. Jacobian conjecture; intersection number; minor root; major root; final root.

Let $\alpha = \sum a_j t^j \in \mathbb{K} \ll t \gg$ and let δ be a real number, we denote $\alpha|_{<\delta} = \sum_{j<\delta} a_j t^j$. If $\beta = \alpha|_{<\delta}$, then we say α is an extension of β . Let $\sigma = \sum_{j<\delta_\sigma} a_j t^j + \pi t^{\delta_\sigma}$ be a π -root, we denote $\sigma|_{\pi=b} = \sum_{j<\delta_\sigma} a_j t^j + b t^{\delta_\sigma}$.

Let $f(x, y)$ and $g(x, y)$ be polynomials in $\mathbb{K}[x, y]$, monic in y . We call (f, g) a Jacobian pair if $f_x g_y - f_y g_x \in \mathbb{K}^*$. Let ξ be a generic element in $\mathbb{K}$ and let $f_\xi = f - \xi$. The following proportional lemma is well-known.

Lemma 2.1. *Let (f, g) be a Jacobian pair. Let $\deg_y f = m$ and $\deg_y g = n$. Let σ be a π -root and let*

$$f_\xi(\sigma) = f_\sigma(\pi) t^{\lambda^f} + \dots, \quad g(\sigma) = g_\sigma(\pi) t^{\lambda^g} + \dots.$$

Then we have

- (i) *If $\lambda^f < 0$ and $\deg_\pi f_\sigma(\pi) > 1$, then $\lambda^f : \lambda^g = m : n = \deg_\pi f_\sigma(\pi) : \deg_\pi g_\sigma(\pi)$.*
- (ii) *If $f_\sigma(\pi)$ has multiple roots, then $f_\sigma(\pi)^n / g_\sigma(\pi)^m \in \mathbb{K}^*$.*

3. ROOT SPLITTING AND $I(f_\xi, f_y)$

In this section, we do not need the Jacobian condition. The results can be applied to other topics.

Lemma 3.1. *Let $p(\pi)$ be a polynomial in $\mathbb{K}[\pi]$ and $p(\pi)$ have e different roots. Then the derivative $p'(\pi)$ has exactly $e - 1$ roots that are not roots of $p(\pi)$.*

Proof. Let $p(\pi) = a(\pi - a_1)^{n_1} \dots (\pi - a_e)^{n_e}$. Then $p'(\pi) = p_1(\pi)(\pi - a_1)^{n_1-1} \dots (\pi - a_e)^{n_e-1}$. It is easy to see that $\deg p_1(\pi) = e - 1$ and that $p_1(\pi)$ and $p(\pi)$ have no common roots. $\square$

Notation 3.2. Let $p(\pi)$ be a polynomial in $\mathbb{K}[\pi]$. Then $e(p(\pi))$ denotes the number of distinct roots of $p(\pi)$.

The following proposition shows the structure of the roots of f_y by split π -roots of f_ξ .

Proposition 3.3. *Let polynomial $f(x, y)$ be monic in y and f_ξ have Puiseux expansion*

$$f_\xi(y) = f_\xi(t^{-1}, y) = \prod_{i=1}^m (y - \alpha_i)$$

over $\mathbb{K} \ll t \gg$ and let $\sigma = \sum_{j<\delta} a_j t^j + \pi t^\delta$ be a π -root of $f_\xi(y)$. Suppose

$$f_\xi(\sigma) = f_\sigma(\pi) t^{\lambda^\sigma} + \text{higher terms in } t$$

and $\deg_\pi f_\sigma(\pi) \geq 1$. Then we have

- (i) $f_y(\sigma) = f'_\sigma(\pi) t^{\lambda^\sigma - \delta} + \text{higher terms in } t$.
- (ii) *If σ is a split π -root of f_ξ , then σ is also a π -root of $f_y(y)$.*
- (iii) *If σ is a split π -root of f_ξ , then there are exactly $e(f_\sigma(\pi)) - 1$ roots $\beta_1, \dots, \beta_{e(f_\sigma(\pi))-1}$ of f_y such that*

$$\text{ord } f_\xi(\beta_i) = \lambda_\sigma \quad \text{for } i = 1, \dots, e(f_\sigma(\pi)) - 1.$$

- (iv) *For any root β of f_y , there is a split π -root σ_β of f_ξ , such that*

$$\text{ord } f_\xi(\beta) = \lambda_{\sigma_\beta}.$$

σ_β and β are said related each other.

(v) Let B be the root set of f_y . For any split π -root σ of f_ξ , let

$$B_\sigma = \{\beta | \beta \in B \text{ and } \text{ord } f_\xi(\beta) = \lambda_\sigma\},$$

then

$$B = \bigsqcup_{\sigma} B_\sigma$$

here σ runs through all split π -root of f_ξ .

Proof. (i) Let $\deg_\pi f_\sigma(\pi) = s$. Reorder index if necessary, we can suppose that

$$\sigma - \alpha_i = (\pi - c_i)t^\delta + \text{higher terms in } t \quad \text{for } i = 1, \dots, s$$

and

$$\sigma - \alpha_i = b_i t^{l_i} + \text{higher terms in } t \quad \text{for } i = s+1, \dots, m$$

here $l_i < \delta$ for $i = s+1, \dots, m$. Then we have

$$\begin{aligned} f_\xi(\sigma) &= \prod_{i=1}^m (\sigma - \alpha_i) \\ &= \prod_{i=1}^s ((\pi - c_i)t^\delta) \prod_{i=s+1}^m b_i t^{l_i} + \dots \\ &= f_\sigma(\pi) t^{\lambda_\sigma} + \dots \end{aligned}$$

As

$$f_y(y) = \sum_{i=1}^m \prod_{j \neq i} (y - \alpha_j) = \sum_{i=1}^m \frac{\prod_{j=1}^m (y - \alpha_j)}{y - \alpha_i}$$

we have

$$\begin{aligned} f_y(\sigma) &= \sum_{i=1}^m \frac{\prod_{j=1}^m (\sigma - \alpha_j)}{\sigma - \alpha_i} \\ &= \sum_{i=1}^s \frac{\prod_{j=1}^m (\sigma - \alpha_j)}{\sigma - \alpha_i} + \dots \\ &= \left(\prod_{i=s+1}^m b_i t^{l_i} \right) \sum_{i=1}^s \frac{\prod_{j=1}^s (\pi - c_j)}{\pi - c_i} t^{(s-1)\delta} + \dots \\ &= f'_\sigma(\pi) t^{\lambda_\sigma - \delta} + \dots \end{aligned}$$

(ii) As $\deg_\pi f_\sigma(\pi) > 1$, therefore $\deg_\pi f'_\sigma(\pi) \geq 1$. By definition, σ is a π -root of $f_y(y)$.

(iii) Let $k = e(f_\sigma(\pi))$ and let $f_\sigma(\pi) = c(\pi - c_1)^{n_1} \dots (\pi - c_k)^{n_k}$. By Lemma 3.1,

$$f'_\sigma(\pi) = f_1(\pi)(\pi - c_1)^{n_1-1} \dots (\pi - c_k)^{n_k-1}$$

with $\deg f_1(\pi) = k - 1$ and also $f_1(\pi)$ and $f_\sigma(\pi)$ have no common roots. Let $f_1(\pi) = b(\pi - b_1) \dots (\pi - b_{k-1})$. By (ii), σ is π -root of f_y . By [Moh83] Proposition 1.2, there are exactly $k - 1$ roots $\beta_1, \dots, \beta_{k-1}$ of f_y such that

$$\beta_i = \sum_{j < \delta} a_j t^j + b_i t^\delta + \dots, \quad \text{ord } f_\xi(\beta_i) = \lambda_\sigma \quad \text{for } i = 1, \dots, k - 1.$$

(iv) As ξ is a generic element of the base field $\mathbb{K}$, $f_\xi(\beta) \neq 0$. Let

$$\delta_\beta = \max\{\text{ord}(\alpha - \beta) | f_\xi(\alpha) = 0\}$$

Write $\beta = \sum b_j t^j$ and let $\sigma_\beta = \sum_{j < \delta_\beta} b_j t^j + \pi t^{\delta_\beta}$, then σ_β is a π -root of f_ξ and $\text{ord } f_\xi(\sigma_\beta) = \text{ord } f_\xi(\beta)$.

(v) This is directly result of (iii) and (iv). $\square$

The intersection number $I(f_\xi, f_y)$ can be expressed by all split π -roots.

Theorem 3.4. *Let $f(x, y)$ be a polynomial in $\mathbb{K}[x, y]$, monic in y . Then*

$$I(f_\xi, f_y) = - \sum_{\sigma} (e(f_\sigma(\pi)) - 1) \lambda_\sigma$$

here σ goes through all splitting π -roots of f_ξ and $f_\xi(\sigma) = f_\sigma(\pi) t^{\lambda_\sigma} + \dots$.

Proof. Let $B = \{\beta | f_y(\beta) = 0\}$. By Proposition 3.3(v), $B = \bigsqcup_{\sigma} B_\sigma$. Therefore

$$I(f_\xi, f_y) = - \sum_{\beta \in B} \text{ord } f_\xi(\beta) = - \sum_{\sigma} \sum_{\beta \in B_\sigma} \text{ord } f_\xi(\beta) = - \sum_{\sigma} (e(f_\sigma(\pi)) - 1) \lambda_\sigma.$$

$\square$

4. $I(f_\xi, f_y)$ AND $I(f_\xi, g)$ AND FINAL MINOR ROOTS

The following lemma is simple but fundamental to our calculations.

Lemma 4.1. *Let $f, g \in \mathbb{K}[x, y]$. Let $f_x g_y - f_y g_x = J(x, y)$. And let α be an element of the Puiseux field $\mathbb{K} \ll t \gg$. Then we have*

$$(4.1) \quad g_y(\alpha) \frac{d}{dt} f(\alpha) - f_y(\alpha) \frac{d}{dt} g(\alpha) = -J(t^{-1}, \alpha) t^{-2}.$$

Proof. By chain rule,

$$(4.2) \quad \frac{d}{dt} f(\alpha) = f_x(\alpha) \frac{dx}{dt} + f_y(\alpha) \frac{d\alpha}{dt}, \quad \frac{d}{dt} g(\alpha) = g_x(\alpha) \frac{dx}{dt} + g_y(\alpha) \frac{d\alpha}{dt}.$$

Substituting (4.2) into LHS of (4.1), simplifying and noting $dx/dt = -t^{-2}$ we get RHS of (4.1). $\square$

From now on, we suppose that (f, g) is a Jacobian pair.

We have the following lemma for final root.

Lemma 4.2. *Let $\alpha = \sum a_j t^j$ be a root of $f_\xi(y)$. Let*

$$\delta = \max\{\text{ord}(\alpha - \beta) \mid g(\beta) = 0\}.$$

And let $\sigma = \sum_{j < \delta} a_j t^j + \pi t^\delta$. Then σ is a final π -root. The final π -root σ and the root α are called related each other.

Proof. Let

$$f_\xi(\sigma) = f_\sigma(\pi) t^{\lambda_f} + \dots, \quad g(\sigma) = g_\sigma(\pi) t^{\lambda_g} + \dots.$$

If σ is not final, then by the proportional lemma (Lemma 2.1), $f_\sigma(\pi)^n / g_\sigma(\pi)^m \in \mathbb{K}^*$. This is a contradiction to the definition of δ . $\square$

Definition 4.3. Let α be a root of $f_\xi(y)$. If $\text{ord } g(\alpha) < 0$ then α is called a major root of $f_\xi(y)$; if $\text{ord } g(\alpha) = 0$ then α is called a minor root of $f_\xi(y)$. When $f(y)$ has two points at infinity with the leading form $(y - a_1 x)^u (y - a_2 x)^v$ and $u < v$, all roots $\alpha = a_1 t^{-1} + \dots$ of $f_\xi(y)$ are minor and the total collection of them are called the principal minor roots of $f_\xi(y)$.

Lemma 4.4. *Let α be a root of $f_\xi(y)$. Let $\sigma = \sum_{j < \delta} a_j t^j + \pi t^\delta$ be the related final π -root. Then we have*

(i) *If α is a major root, then $\delta < 1$*

(ii) If α is a minor root, then $\delta > 1$.

Proof. Let

$$f_\xi(\sigma) = f_\sigma(\pi)t^{\lambda^f} + f_1(\pi)t^{\lambda^f+\varepsilon} + \dots, \quad g(\sigma) = g_\sigma(\pi)t^{\lambda^g} + g_1(\pi)t^{\lambda^g+\varepsilon} + \dots$$

here $\varepsilon > 0$.

(i) Since α is major, then $\lambda^f < 0$ and $\lambda^g < 0$. By [Moh83] Proposition 4.1,

$$(\lambda^f f_\sigma(\pi)g'_\sigma(\pi) - \lambda^g f'_\sigma(\pi)g_\sigma(\pi))t^{\lambda^f+\lambda^g-1} + \dots = \frac{\partial(f(\sigma), g(\sigma))}{\partial(t, \pi)} = -Jt^{\delta-2}.$$

Since σ is final, therefore $\lambda^f f_\sigma(\pi)g'_\sigma(\pi) - \lambda^g f'_\sigma(\pi)g_\sigma(\pi) \neq 0$. Hence $\lambda^f + \lambda^g - 1 = \delta - 2$. Therefore $\delta < 1$.

(ii) Now $\lambda^f = 0 = \lambda^g$. We have

$$(\varepsilon f_1(\pi)g'_\sigma(\pi) - \varepsilon f'_\sigma(\pi)g_1(\pi))t^{\varepsilon-1} + \dots = \frac{\partial(f(\sigma), g(\sigma))}{\partial(t, \pi)} = -Jt^{\delta-2}.$$

Hence $\varepsilon - 1 \leq \delta - 2$. Therefore $\delta > 1$. $\square$

Now we can show the relationship between our minor roots and [Moh83] minor discs of a split π -root.

Corollary 4.5. Let $\tau = \sum_{j < \delta_r} a_j t^j + c_r t^{\delta_r}$ be a minor disc center defined as in [Moh83] Proposition 6.1. Let

$$E_\tau^f = \{\alpha \mid f_\xi(\alpha) = 0 \text{ and } \text{ord}(\tau - \alpha) > \delta_r\}.$$

Then all elements in E_τ^f are minor roots.

Proof. Let $\alpha \in E_\tau^f$, and let

$$\delta^* = \min\{\text{ord}(\alpha - \beta) \mid \beta \text{ is root of } f_\xi(y)g(y) \text{ and } \text{ord}(\tau - \beta) > \delta_r\}.$$

Then by [Moh83] Proposition 6.1, $\delta^* \geq 1$. According to Lemma 4.4, α is a minor root. $\square$

Definition 4.6. Let $\sigma = \sum_{j < \delta_\sigma} a_j t^j + \pi t^{\delta_\sigma}$ be a final π -root of f_ξ . Let polynomial $h(y) \in \mathbb{K}[x, y]$. We define set

$$D_\sigma^h = \{\alpha \mid h(\alpha) = 0 \text{ and } \text{ord}(\sigma - \alpha) = \delta_\sigma\}.$$

If $h(\alpha) = 0$ and $\alpha \in D_\sigma^h$, we call α and σ are related each other. We denote $|D_\sigma^h|$ be the cardinal of set D_σ^h . We define sets

$$P_m = \{\sigma \mid \sigma \text{ is final minor } \pi\text{-root of } f_\xi\},$$

$$P_M = \{\sigma \mid \sigma \text{ is final major } \pi\text{-root of } f_\xi\}.$$

Intersection numbers $I(f_\xi, g)$ and $I(f_\xi, f_y)$ have relationships with final minor π -roots.

Theorem 4.7. Let (f, g) be a Jacobian pair. Then we have

(i)

$$I(f_\xi, f_y) \leq \deg_y f - 1 + \sum_{\sigma \in P_m} (|D_\sigma^{f_\xi}| - 1)(\delta_\sigma - 1)$$

(ii)

$$I(f_\xi, g) \geq 1 + \sum_{\sigma \in P_m} (\delta_\sigma - 1).$$

If there are no final minor roots, then

(iii)

$$I(f, f_y) = \deg_y f - 1$$

$$I(f, g) = 1.$$

Proof. Let $f_\xi(\alpha) = 0$. By Lemma 4.1,

$$-f_y(\alpha) \frac{d}{dt} g(\alpha) = g_y(\alpha) \frac{d}{dt} f(\alpha) - f_y(\alpha) \frac{d}{dt} g(\alpha) = -Jt^{-2}.$$

If α is a major root, then $\text{ord } g(\alpha) < 0$. Therefore

$$\text{ord } \frac{d}{dt} g(\alpha) = \text{ord } g(\alpha) - 1.$$

Thus

$$\text{ord } f_y(\alpha) g(\alpha) = -1.$$

If α is a minor root, then $\text{ord } g(\alpha) = 0$. Define

$$\delta = \max\{\text{ord}(\beta - \alpha) \mid g(\beta) = 0\}.$$

By Proposition 3.3, $\text{ord } f_y(\alpha) = -\delta$. Therefore

$$\text{ord } f_y(\alpha) g(\alpha) = -\delta = -1 - (\delta - 1).$$

Thus

$$\begin{aligned} (4.3) \quad I(f_\xi, f_y g) &= -\text{ord} \prod_{\{\alpha \mid f_\xi(\alpha)=0\}} f_y(\alpha) g(\alpha) \\ &= -\sum_{\sigma \in P_m} \sum_{\alpha \in D_\sigma^{f_\xi}} \text{ord } f_y(\alpha) g(\alpha) - \sum_{\sigma \in P_M} \sum_{\alpha \in D_\sigma^{f_\xi}} \text{ord } f_y(\alpha) g(\alpha) \\ &= \sum_{\sigma \in P_m} |D_\sigma^{f_\xi}| (1 + (\delta_\sigma - 1)) + \sum_{\sigma \in P_M} |D_\sigma^{f_\xi}| \\ &= \deg_y f + \sum_{\sigma \in P_m} |D_\sigma^{f_\xi}| (\delta_\sigma - 1). \end{aligned}$$

Now let $f_y(\beta) = 0$. Note that $\text{ord } f_\xi(\beta) = 0$ if and if $\beta \in D_\sigma^{f_y}$ for some $\sigma \in P_m$ and $\text{ord } g_y(\beta) \geq -\delta_\sigma$. By Proposition 3.3, $|D_\sigma^{f_y}| = |D_\sigma^{f_\xi}| - 1$. With the similar reasons as above, we have

$$\begin{aligned} (4.4) \quad I(f_y, f_\xi g_y) &= -\text{ord} \prod_{\{\beta \mid f_y(\beta)=0\}} f_\xi(\beta) g_y(\beta) \\ &\leq \deg_y f_y + \sum_{\sigma \in P_m} |D_\sigma^{f_y}| (\delta_\sigma - 1) \\ &= \deg_y f - 1 + \sum_{\sigma \in P_m} (|D_\sigma^{f_\xi}| - 1) (\delta_\sigma - 1). \end{aligned}$$

As $I(f_y, f_\xi g_y) = I(f_y, f_\xi) + I(f_y, g_y)$, and from the Jacobian condition, $I(f_y, g_y) = 0$, (i) is proved. Note $I(f_\xi, f_y g) = I(f_\xi, f_y) + I(f_\xi, g)$ and subtract (4.4) from (4.3), (ii) is proved. If there are no final minor roots, then all above inequalities become equal and (iii) follows. $\square$

Corollary 4.8. *Let (f, g) be a Jacobain pair. The following are equivalent.*

- (i) $\mathbb{K}(f, g) = \mathbb{K}(x, y)$.
- (ii) $\mathbb{K}[f, g] = \mathbb{K}[x, y]$.
- (iii) $I(f, g) = 1$.
- (iv) f has no minor roots.
- (v) $I(f, f_y) = \deg_y f - 1$.

Proof. Let ξ and η be symbols, let $f_\xi = f - \xi$, $g_\eta = g - \eta$. And let $F(\xi, \eta, x) = \text{Res}_y(f_\xi, g_\eta)$. Then $F(\xi, \eta, x)$ is irreducible ([Moh83] page 150) and $F(f(x, y), g(x, y), x) = 0$. So $F(f, g, X)$ is the define equation of x over $\mathbb{K}(f, g)$. Thus $I(f_\xi, g_\mu)$ is the degree of field extension $[\mathbb{K}(x, y) : \mathbb{K}(f, g)]$, which is equal to 1 if and only if there is no minor root from above theorem. We also see from the proof of the last theorem, if there is no minor root, then the coefficient of the leading term of $F(\xi, \mu, x)$ in x is in $\mathbb{K}^*$. The corollary is thus proved. $\square$

Corollary 4.9. ([Xu93]) *Let $f(x, y)$ be a polynomial, monic in y . If f is an embedded line, then $I(f, f_y) = \deg_y f - 1$. In other words, the curve f has $\deg_y f - 1$ vertical tangent lines.*

Proof. By Abhyankar-Moh's Epimorphism Theorem ([AM75]), there exists polynomial $g(x, y)$ such that Jacobian $J(f, g) \in \mathbb{K}^*$ and $\mathbb{K}[x, y] = \mathbb{K}[f, g]$. By above corollary, $I(f, f_y) = \deg_y f - 1$. $\square$

5. $I(f_\xi, g)$ EXPRESSED BY FINAL MAJOR ROOTS

Now we express $I(f_\xi, g)$ with final major π -roots. We define

$$\begin{aligned} I_M(f, g) &= - \sum_{\sigma \in P_M} |D_\sigma^f| \lambda_\sigma^g \\ &= \frac{\deg_y g}{\deg_y f + \deg_y g} \sum_{\sigma \in P_M} |D_\sigma^f| (1 - \delta_\sigma). \end{aligned}$$

Theorem 5.1. *Let (f, g) be a Jacobian pair. Then we have*

$$I(f_\xi, g) = I_M(f, g)$$

Proof. Let $\deg_y f = m$ and $\deg_y g = n$. Let $f_\xi(\alpha) = 0$. If α is a major root, then $\text{ord } g(\alpha) < 0$. Let σ be its final π -root. And let

$$f(\sigma) = f_\sigma(\pi) t^{\lambda_\sigma^f} + \cdots, \quad g(\sigma) = g_\sigma(\pi) t^{\lambda_\sigma^g} + \cdots.$$

By [Moh83] Proposition 4.6,

$$-\lambda_\sigma^g = n \frac{1 - \delta_\sigma}{n - M_1} = \frac{n}{m + n} (1 - \delta_\sigma).$$

If α is a minor root, then $\text{ord } g(\alpha) = 0$. Therefore

$$\begin{aligned} I(f_\xi, g) &= - \text{ord} \prod_{\{\alpha | f_\xi(\alpha)=0\}} g(\alpha) \\ &= - \sum_{\sigma \in P_M} \sum_{\alpha \in D_\sigma^f} \text{ord } g(\alpha) - \sum_{\sigma \in P_m} \sum_{\alpha \in D_\sigma^{f_\xi}} \text{ord } g(\alpha) \\ &= - \sum_{\sigma \in P_M} |D_\sigma^f| \lambda_\sigma^g \\ &= \frac{n}{m + n} \sum_{\sigma \in P_M} |D_\sigma^f| (1 - \delta_\sigma). \end{aligned}$$

$\square$

Zhang [Zha91] gives estimate for the degree of the field extension $[\mathbb{K}(x, y) : \mathbb{K}(f, g)] \leq \min(\deg f, \deg g)$. Using the theorem above, we can give a better estimate.

Corollary 5.2. *Let (f, g) be a Jacobian pair. Let $\deg_y f > 1$ and $\deg_y g > 1$. Then the field extension degree $[\mathbb{K}(x, y) : \mathbb{K}(f, g)] < \frac{\deg_y f \deg_y g}{\deg_y f + \deg_y g}$.*

Proof. We only need note that in the proof of the last theorem, $0 < \delta_\sigma < 1$ ([Moh83] Proposition 5.3) and

$$\sum_{\sigma \in P_M} |D_\sigma^f| (1 - \delta_\sigma) < \sum_{\sigma \in P_M} |D_\sigma^f| \leq \deg_y f.$$

□

We define

$$I_m(f, g) = 1 + \sum_{\sigma \in P_m} (\delta_\sigma - 1).$$

By combining Theorem 4.7 and Theorem 5.1, we have

Corollary 5.3. *Let (f, g) be a Jacobian pair. Then*

$$I_M(f, g) \geq I_m(f, g).$$

6. APPLICATIONS

We apply Corollary 5.3 to special cases. We show that these formulas work in some cases but are not effect in other cases.

6.1. Case (75, 50). We take case (75, 50) as an example. According to [Moh83], page 202, there are two sub cases [For (75, 50) second sub case, $\delta_1 = 1/3$ should be $\delta_1 = 2/3$]. We only show the second one here, as it provides more splitting possibilities, therefore more complicated. i.e. We have

$$n = 75, \quad m = 50, \quad M_2 = 55, \quad M_3 = 73, \quad V_3 = 4, \quad V_2 = 2, \quad \delta_2 = 1/5, \quad \delta_1 = 2/3.$$

Let σ_2 be the major π -root with the split order δ_2 . There are two possibilities of splittings for σ_2 .

(i)

$$f(\sigma_2) = a((\pi^5 - c_1)(\pi^5 - c_2))^4 t^{-2} + \dots, \quad g(\sigma_2) = b((\pi^5 - c_1)(\pi^5 - c_2))^6 t^{-3} + \dots.$$

All these 10 roots of $f_{\sigma_2}(\pi)$ and $g_{\sigma_2}(\pi)$ are major. Each of them extends to final major π -roots σ_{1i} with split order δ_1 :

$$f(\sigma_{1i}) = a_{0i}\pi(\pi^3 - a_{1i})t^{-2/15} + \dots, \quad g(\sigma_{1i}) = b_{0i}(\pi^3 - b_{1i})(\pi^3 - b_{2i})t^{-1/5} + \dots$$

for $i = 1, \dots, 10$. We have only one final minor π -root with split order $V_3/U_3 = 4$. Therefore, from final minor roots, we have $I_m(f, g) = 1 + (4 - 1) = 4$, while final major roots give

$$I_M(f, g) = 10 \times 4 \times 1/5 = 8.$$

They do not work in this case.

(ii)

$$f(\sigma_2) = a((\pi^5 - c_1)^2(\pi^5 - c_2)(\pi^5 - c_3))^2 t^{-2} + \dots, \\ g(\sigma_2) = b((\pi^5 - c_1)^2(\pi^5 - c_2)(\pi^5 - c_3))^3 t^{-3} + \dots.$$

In this case, 5 roots from $(\pi^5 - c_1)$ are major and extend to final major π -roots σ_{1i} with split order δ_1 :

$$f(\sigma_{1i}) = a_{0i}\pi(\pi^3 - a_{1i})t^{-2/15} + \dots, \quad g(\sigma_{1i}) = b_{0i}(\pi^3 - b_{1i})(\pi^3 - b_{2i})t^{-1/5} + \dots$$

for $i = 1, \dots, 5$; while 10 roots from $(\pi^5 - c_2)(\pi^5 - c_3)$ are minor and extend to final minor π -roots with split order $6/5$. So we have

$$I_m(f, g) = 1 + (4 - 1) + 10 \times (6/5 - 1) = 6 \text{ from final minor roots while}$$

$$I_M(f, g) = 5 \times 4 \times 1/5 = 4.$$

This is a contradiction and we can rule out this case.

6.2. **Case (84, 56).** From [Moh83] page 202, there are two cases.

(i)

$$n = 84, \quad m = 56, \quad M_2 = 64, \quad M_3 = 82, \quad V_3 = 3, \quad V_2 = 2, \quad \delta_2 = 2/7, \quad \delta_1 = 16/21.$$

Let σ_2 be the major π -root with the split order δ_2 .

$$f(\sigma_2) = a((\pi^7 - c_1)^2(\pi^7 - c_2))^2 t^{-2} + \dots, \quad g(\sigma_2) = b((\pi^7 - c_1)^2(\pi^7 - c_2))^3 t^{-3} + \dots.$$

All that 7 roots of $(\pi^7 - c_1)$ are major, each of them extends to a final major π -root σ_{1i} with split order δ_1 :

$$f(\sigma_{1i}) = a_{0i}\pi(\pi^3 - a_{1i})t^{-2/21} + \dots, \quad g(\sigma_{1i}) = b_{0i}(\pi^3 - b_{1i})(\pi^3 - b_{2i})t^{-3/21} + \dots$$

for $i = 1, \dots, 7$.

All that 7 roots of $(\pi^7 - c_2)$ are minor, each of them extends to a final minor π -root with order $9/7$. We also have one principal final minor π -root with split order $V_3/U_3 = 3$. Therefore, from final minor roots, we have $I_m(f, g) = 1 + 7 \times (9/7 - 1) + (3 - 1) = 5$, while final major roots give $I_M(f, g) = 4 \times 7 \times 3/21 = 4$, and that is a contradiction. So we can rule out this case.

(ii)

$$n = 84, \quad m = 56, \quad M_2 = 72, \quad M_3 = 82, \quad V_3 = 3, \quad V_2 = 5, \quad \delta_2 = 1/4, \quad \delta_1 = 7/12.$$

$$f(\sigma_2) = a((\pi^4 - c_1)^5 \pi)^2 t^{-14/4} + \dots, \quad g(\sigma_2) = b((\pi^4 - c_1)^5 \pi)^3 t^{-21/4} + \dots.$$

In this case, 4 roots from $(\pi^4 - c_1)$ are major and extend to final major π -roots σ_{1i} with split order δ_1 :

$$f(\sigma_{1i}) = a_{0i}\pi(\pi^3 - a_{1i})(\pi^3 - a_{2i})(\pi^3 - a_{3i})t^{-2/12} + \dots,$$

$$g(\sigma_{1i}) = b_{0i}(\pi^3 - b_{1i})(\pi^3 - b_{2i})(\pi^3 - b_{3i})(\pi^3 - b_{4i})(\pi^3 - b_{5i})t^{-3/12} + \dots$$

for $i = 1, \dots, 4$; while 1 root from $(\pi - 0)$ is minor and extend to final minor π -root with split order 2. So we have $I_m(f, g) = 1 + (2 - 1) + (3 - 1) = 4$ from final minor roots while $I_M(f, g) = 4 \times 10 \times 3/12 = 10$. We can not rule out this case.

7. SPLITTING OF THE PRINCIPAL MINOR ROOTS

In this section we will show that the principal minor roots are not splitting at order 1 for all effective quasi-approximate roots. We need a few preliminaries first.

7.1. Planar semigroups. For Abhyankar-Moh planar semigroup theory and characteristic δ -sequence, we refer [AM75], [SS94] and [Xu14]. We recall definitions of q -sequence and M -sequence

$$M_1 = -\delta_1$$

$$q_i = \delta_{i-1} \frac{d_{i-1}}{d_i} - \delta_i, \quad M_i = M_{i-1} + q_i$$

for $i = 2, \dots, h$.

For an integer subset A , $\Gamma(A)$ denotes the semigroup generated by all elements in A .

Lemma 7.1. *Let $\delta = (\delta_0, \dots, \delta_h)$ ($h \geq 2$) be a characteristic δ -sequence and k be an integer with $2 \leq k \leq h$. Then we have*

- (i) $\delta_k + M_k \in \Gamma(\delta_1, \dots, \delta_{k-1})$,
- (ii) $\delta_k + M_k - \delta_0 \notin \Gamma(\delta_0, \dots, \delta_{k-1})$.

Proof. $\delta_1 + M_1 = 0$. As

$$\delta_k + M_k = \delta_{k-1} \frac{d_{k-1}}{d_k} - q_k + M_k = \delta_{k-1} \left(\frac{d_{k-1}}{d_k} - 1 \right) + \delta_{k-1} + M_{k-1}$$

by induction, we have

$$\delta_k + M_k = \delta_{k-1} \left(\frac{d_{k-1}}{d_k} - 1 \right) + \dots + \delta_1 \left(\frac{d_1}{d_2} - 1 \right)$$

(i) is proved. Now

$$\delta_k + M_k - \delta_0 = \delta_{k-1} \left(\frac{d_{k-1}}{d_k} - 1 \right) + \dots + \delta_1 \left(\frac{d_1}{d_2} - 1 \right) - \delta_0$$

is a standard expansion by the δ -sequence, by [Xu14] Lemma 2.3, (ii) is proved. $\square$

7.2. One kind of special ordinary differential equations of polynomials. After analyzing the proof of the second part of [Moh83] Proposition A.2, we have

Lemma 7.2. *Let $\deg p(\pi) = m$ and $\deg q(\pi) = (l-1)m+1$. Suppose $p(\pi), q(\pi)$ satisfy the following equation*

$$D(m, m(l-1), p(\pi), q(\pi)) = cp(\pi)^l$$

where $c \in \mathbb{K}^*$. Then $q(\pi) = \frac{c}{m}(\pi - a)p(\pi)^{l-1}$ for some $a \in \mathbb{K}$.

7.3. Quasi-approximate roots of a polynomial curve. Let (f, g) be a Jacobian pair, monic in y and f has two points at infinity. Let $\deg f = m$ and $\deg g = n$. Consider y as a parameter, (f, g) defines a polynomial curve. From [Moh83], there are effective index $s > 2$ and quasi-approximate roots $T_i(f, g) \in \mathbb{K}[f, g]$ with $T_0 = g$, $T_1 = f$ and $\deg T_i = -\mu_i$ for $i = 0, \dots, s$.

Let $d_i = \gcd(-\mu_0, \dots, -\mu_{i-1})$ for $i = 1, \dots, s+1$. Then $(-\mu_0/d_{s+1}, \dots, -\mu_s/d_{s+1})$ is a δ -sequence. Let the multiplicity of the principal minor roots of f is $m \frac{u_s}{d_s}$, here $u_s < v_s = d_s - u_s$. Then the multiplicities of the principal minor roots of $T_0, \dots, T_{s-1}, T_s$ are

$$-\mu_0 \frac{u_s}{d_s}, \dots, -\mu_{s-1} \frac{u_s}{d_s}, (-\mu_s - 2) \frac{u_s}{d_s} + 1$$

respectively.

Proposition 7.3. *Let σ_1 be the π -root of order 1 for the principal minor roots of f . Let*

$$\begin{aligned} T_0(\sigma_1) &= T_{0,\sigma_1}(\pi) t^{\frac{v_s - u_s}{d_s} \mu_0} + \dots \\ &\vdots \\ T_{s-1}(\sigma_1) &= T_{s-1,\sigma_1}(\pi) t^{\frac{v_s - u_s}{d_s} \mu_{s-1}} + \dots \\ T_s(\sigma_1) &= T_{s,\sigma_1}(\pi) t^{\frac{v_s - u_s}{d_s} (\mu_s + 2)} + \dots \end{aligned}$$

Then $T_{0,\sigma_1}(\pi), \dots, T_{s,\sigma_1}(\pi)$ are powers of a common linear polynomial.

Proof. Using the proof of [Moh83] Proposition 4.6 verbatim, we get

$$T_{0,\sigma_1}(\pi) = C_0 p(\pi)^{\frac{-\mu_0}{d_s}}, \dots, T_{s-1,\sigma_1}(\pi) = C_{s-1} p(\pi)^{\frac{-\mu_{s-1}}{d_s}}$$

where $\deg p(\pi) = u_s$.

Finally by Lemma 7.2, we have

$$T_{s,\sigma_1}(\pi) = C_s p(\pi)^{\frac{-\mu_s - 2}{d_s}} (\pi - a)$$

where $a \in \mathbb{K}$, $C_i \in \mathbb{K}^*$ for $i = 0, \dots, s$.

We claim $p(\pi) = (\pi - a)^{u_s}$. If not, let $\pi = b$ be a root of $p(\pi)$ and $b \neq a$. We extend $\sigma_1|_{\pi=b}$ to a final minor π -root σ_0 with split order δ_0 .

First we point out that for any $1 < \delta < \delta_0$ and π -root $\sigma = \sigma_0|_{<\delta} + \pi t^\delta$ we have that for some $\lambda < 0$,

$$\begin{aligned} \text{ord } T_0(\sigma) &= (-\mu_0)\lambda \\ &\vdots \\ \text{ord } T_{s-1}(\sigma) &= (-\mu_{s-1})\lambda \\ \text{ord } T_s(\sigma) &= (-\mu_s - 2)\lambda \end{aligned}$$

Using the proof of [Moh83] Proposition 4.2 verbatim, we conclude that σ is a distribution detector for $T_0(y), \dots, T_s(y)$.

Hence we have

$$\begin{aligned} T_0(\sigma_0) &= T_{0,\sigma_0}(\pi)t^0 + \dots \\ &\vdots \\ T_{s-1}(\sigma_0) &= T_{s-1,\sigma_0}(\pi)t^0 + \dots \\ T_s(\sigma_0) &= T_{s,\sigma_0}(\pi)t^0 + \dots \end{aligned}$$

where

$$\begin{aligned} \deg T_{0,\sigma_0}(\pi) &= \frac{-\mu_0}{d_s}r \\ &\vdots \\ \deg T_{s-1,\sigma_0}(\pi) &= \frac{-\mu_{s-1}}{d_s}r \\ \deg T_{s,\sigma_0} &= \frac{-\mu_s - 2}{d_s}r \end{aligned}$$

for some positive integer r and r satisfies $1 \leq r \leq u_s$.

We consider polynomial curve $(f_{\sigma_0}(\pi), g_{\sigma_0}(\pi))$ with π as the parameter. Then $\{T_{i,\sigma_0}(\pi)\}_{0 \leq i \leq s-1}$ are quasi-approximate roots of the curve.

$$T_{s,\sigma_0}(\pi) = T_s(f_{\sigma_0}(\pi), g_{\sigma_0}(\pi)) \in \mathbb{K}[f_{\sigma_0}(\pi), g_{\sigma_0}(\pi)]$$

As $\gcd(\frac{-\mu_0}{d_s}r, \dots, \frac{-\mu_{s-1}}{d_s}r) = r$ and r divides $\deg T_{s,\sigma_0}(\pi) = \frac{-\mu_s - 2}{d_s}r$, we conclude that $\frac{-\mu_s - 2}{d_s}r$ is in the semigroup $\Gamma(\frac{-\mu_0}{d_s}r, \dots, \frac{-\mu_{s-1}}{d_s}r)$.

On the other hand,

$$-\mu_s - 2 = -\mu_s + M_s - n \notin \Gamma(-\mu_0, \dots, -\mu_{s-1})$$

by Lemma 7.1. This is a contradiction. $\square$

Remark 7.4. By [ML14] the coefficient of t in any root α of $f(y)$ is zero. We do not find this information can simplify the proof that T_s is not split at order 1.

Corollary 7.5. Suppose $u_s > 1$ and let σ be a π -root of f for the principle minor roots. If order $\delta_\sigma < \frac{v_s + 1}{u_s + 1}$ then σ can not split to u_s different roots.

Proof. Suppose σ split to u_s different roots. For simplify we denote $\delta_\sigma = \delta$. First from last Proposition we know $\delta > 1$. We can write

$$f(\sigma) = a_0 p(\pi)^{m/d_s} t^{-v_s \frac{m}{d_s} + u_s \frac{m}{d_s} \delta} + \dots$$

here

$$p(\pi) = (\pi - c_1) \cdots (\pi - c_{u_s})$$

with

$$c_i \neq c_j \text{ if } i \neq j$$

From above Proposition, we know that $\delta > 1$ and we can write

$$g(\sigma) = a_1 p(\pi)^{n/d_s} t^{-v_s \frac{n}{d_s} + u_s \frac{n}{d_s} \delta} + \dots$$

$$(T_s)_f(\sigma) = b p(\pi)^{(-\mu_s + n - 2)/d_s} t^{-v_s \frac{-\mu_s + n - 2}{d_s} + u_s \frac{-\mu_s + n - 2}{d_s} \delta} + \dots$$

$$T_s(\sigma) = a_s q(\pi) t^{-v_s \frac{-\mu_s + n - 2}{d_s} - 1 + (u_s \frac{-\mu_s + n - 2}{d_s} + 1) \delta} + \dots$$

here $\deg q(\pi) = u_s \frac{-\mu_s + n - 2}{d_s} + 1$. By [Moh83] [Proposition 4.1], we have

$$(7.1) \quad \frac{\partial(T_s(\sigma), g(\sigma))}{\partial(t, \pi)} = -J \cdot (T_s)_f(\sigma) t^{-2+\delta}$$

With the above information, and note that

$$-v_s \frac{-\mu_s + n - 2}{d_s} - 1 + (u_s \frac{-\mu_s + n - 2}{d_s} + 1) \delta < -v_s - 1 + (u_s + 1) \delta < 0,$$

so every root of $p(\pi)$ is a root of $q(\pi)$ and $p(\pi)$ has no multi roots, we have $q(\pi) = p(\pi) q_1(\pi)$. We can use same reason to $q_1(\pi)$ and on. Finally we have

$$q(\pi) = p(\pi)^{\frac{-\mu_s + n - 2}{d_s}} (\pi - c)$$

for some c . the above equation 7.1 can be simplified to

$$\frac{\partial}{\partial(t, \pi)} (p(\pi) t^{-v_s + u_s}, (\pi - c) t^{-1+\delta}) = C p(\pi) t^{-v_s + u_s - 2 + \delta}.$$

But this equation can not hold and it is a contradiction. $\square$

8. CASE (99, 66) AND SPLIT OF PRINCIPAL MINOR ROOTS

Moh writes in [Moh83] page 209 on the principal minor roots, “Namely, there is an [sic] unique π -root σ of $g(y)$ of the following form

$$\sigma = a_{-1} t^{-1} + a_0 + a_1 t + \pi t^2$$

with

$$g(\sigma) = g_\pi(\sigma) t^{-18} + \dots \quad [\text{sic, } g_\pi(\sigma) \text{ should be } g_\sigma(\pi)]$$

The polynomial $g_\sigma(\pi)$ is either a power of a linear polynomial or the 9-th power of a cubic polynomial with precisely two roots.”

For a long time, we try to give judgement for this statement. To do so, there are three claims have to be proven.

- (i) The principal minor roots do not split except at π -root order $\delta = 2$.
- (ii) $g_\sigma(\pi)$ can not have three roots when $\delta = 2$.
- (iii) Let $\delta = 2$ and let $g_\sigma(\pi) = c(\pi - b_1)^{18}(\pi - b_2)^9$. Roots in the disc with the center $\sigma|_{\pi=b_1}$ do not split before final.

We can not find materials to support these claims in his paper. We sent Moh an email to point out the possible gaps. He replied with, “I will make an investigation of the issue and reply to your e-mail as soon as possible.” on Jan. 06, 2016.

We show that (ii) and (iii) are true, but there is one exception case for (i) open.

Our tools are two constrains: one is that the denominator of order $\delta \leq u_s = 3$; another is equation 7.1.

From [Moh83] page 202, we have data

$$(8.1) \quad n = 99, \quad m = 66, \quad M_2 = 77, \quad M_3 = 97, \quad V_3 = 8, \quad V_2 = 8, \quad \delta_2 = 1/3, \quad \delta_1 = 4/9.$$

Let σ be π -root of the principal minor roots. From data 8.1 we get that $-\mu_2 = 55$ and $-\mu_3 = 145$. We have

$$\begin{aligned} f(\sigma) &= p(\pi)^6 t^{6(-8+3\delta)} + \dots \\ g(\sigma) &= p(\pi)^9 t^{9(-8+3\delta)} + \dots \\ T_2(\sigma) &= p(\pi)^5 t^{5(-8+3\delta)} + \dots \\ (T_3)_f(\sigma) &= p(\pi)^{22} t^{22(-8+3\delta)} + \dots \\ T_3(\sigma) &= q(\pi) t^{13(-8+3\delta)-1+\delta} + \dots \end{aligned}$$

here $\deg p(\pi) = u_3 = 3$ and $\deg q(\pi) = 13 * 3 + 1$.

(i) By the constrain of the denominator of δ , if δ splits, it must split to 2 or 3 roots.

When $\delta \neq 2$ and $\delta \neq \frac{5}{2}$. From equation 7.1, we can step by step deduce

$$q(\pi) = p(\pi)^{13}(\pi - c)$$

and the equation 7.1 can not hold for it.

When $\delta = 2$. As there is equation

$$(8.2) \quad \frac{\partial}{\partial(t, \pi)}(\pi(\pi + 3a)^2(\pi - 2a)t^{-1}, \pi^2(\pi + 3a)t^{-2}) = 5\pi^4(\pi + 3a)^2 t^{-4},$$

The equation 7.1 produces Moh's 2 roots split case.

When $\delta = \frac{5}{2}$. From equation 7.1, we can deduce

$$q(\pi) = p(\pi)^{10} q_1(\pi)$$

here $\deg q_1(\pi) = 3 * 3 + 1$. The equation 7.1 gives us

$$\frac{\partial}{\partial(t, \pi)}(q_1(\pi), p(\pi)t^{-\frac{1}{2}}) = -p(\pi)^4 t^{-\frac{3}{2}}.$$

But this can hold for any $p(\pi)$ and $q_1(\pi) = -2 \int p(\pi)^3 dx$. So there is possible $p(\pi) = \pi(\pi^2 - c)$. Thus this case is open and it suggests case (99, 66) is open.

(ii) This case is proved by Corollary 7.5.

(iii) After π -root extends to $\delta > 2$ and $\delta < 3$ along $\sigma|_{\pi=b_1}$, we have

$$\begin{aligned} f(\sigma) &= p(\pi)^6 t^{12(-3+\delta)} + \dots \\ g(\sigma) &= p(\pi)^9 t^{18(-3+\delta)} + \dots \\ T_2(\sigma) &= p(\pi)^5 t^{10(-3+\delta)} + \dots \\ T_3(\sigma) &= q(\pi) t^{25(-3+\delta)} + \dots \end{aligned}$$

here $\deg p(\pi) = 2$, $\deg q(\pi) = 25$ and $\frac{\partial}{\partial(t, \pi)}(q(\pi)t^{25(-3+\delta)}, p(\pi)^9 t^{18(-3+\delta)}) = 0$.

As $\gcd(12, 18, 10, 25) = 1$, $p(\pi)$ and $q(\pi)$ are powers of a common linear factor $(\pi - c)$, i.e. π -root is not split at this order δ . The claim is proved.

ACKNOWLEDGEMENT

The author thanks Professor Mattias Josson and Professor Christian Valqui for pointing out errors in previous version.

REFERENCES

- [AM75] S.S. Abhyankar and T.T. Moh, *Embeddings of the line in a plane*, J. Reine Angew. Math. **276** (1975), 148–166. [↑4](#), [7.1](#)
- [ML14] L. Makar-Limanov, *On the Newton polygon of a Jacobian mate*, Automorphisms in birational and affine geometry, 469476, Springer Proc. Math. Stat., 79, Springer, Cham, (2014). [↑7.4](#)
- [Moh83] T.T. Moh, *On the Jacobian conjecture and configurations of roots*, J. Reine Angew. Math. **340** (1983), 140–212. MR691964 (84m:14018) [↑1](#), [2](#), [3](#), [4](#), [4.5](#), [4](#), [4](#), [5](#), [5](#), [6.1](#), [6.2](#), [7.2](#), [7.3](#), [7.3](#), [7.3](#), [8](#), [8](#)
- [SS94] A. Sathaye and J. Stenerson, *Plane Polynomial Curves*, Algebraic Geometry and its Applications (C.L. Bajaj, ed), Springer-Verlag, New York-Berlin-Heidelberg (1994), 121–142. [↑7.1](#)
- [Xu93] Yansong Xu, *On the Jacobian conjecture and affine lines*, 1993. Purdue University, Ph.D thesis. [↑4.9](#)
- [Xu14] ———, *A strong Abhyankar-Moh theorem and criterion of embedded line*, J. of Algebra **409** (2014), 382–386. [↑7.1](#), [7.1](#)
- [Zha91] Yitang Zhang, *The Jacobian conjecture and the degree of field extension*, 1991. Purdue University, Ph.D thesis. [↑5](#)

Email address: `yansong_xu@yahoo.com`